

CUSTOMER SEGMENTATION & CHURN PATTERN ANALYTICS IN EUROPEAN BANKING

Business Analytics Internship Project

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Project: Customer Segmentation & Churn Pattern Analytics in European Banking

Internship Project Report

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EXECUTIVE SUMMARY

Customer churn represents a significant business challenge for banking institutions, as customer attrition can affect customer lifetime value, acquisition efficiency, and long-term customer relationships. This project, titled “Customer Segmentation & Churn Pattern Analytics in European Banking,” was undertaken to identify customer segments with relatively higher churn risk and translate the observed patterns into actionable business insights.

The study uses customer-level banking data containing demographic, geographic, financial, tenure, product-related, and engagement characteristics. A structured Business Analytics approach was followed involving data understanding, validation and preparation, customer segmentation, exploratory data analysis, churn-rate analysis, visualization, and business interpretation. Customers were analysed across key segmentation dimensions including geography, age, credit score, tenure, account balance, and activity status. An interactive Tableau dashboard was developed to support visual comparison and exploration of churn patterns.

The analysis identified an overall customer churn rate of 20.4%. Significant differences were observed across several customer segments. Germany recorded the highest geographical churn rate at 32.44%, compared with Spain at 16.67% and France at 16.15%. Age emerged as one of the most distinctive segmentation dimensions, with customers aged 46–60 recording a churn rate of 51.1%, substantially above the overall churn level.

Customer engagement also demonstrated an important relationship with churn. Inactive customers recorded a churn rate of 26.85%, compared with 14.27% among active customers. Balance-based analysis showed churn rates of 24.11% among Minimum/Low Balance customers and 23.12% among High Balance customers. New customers recorded a churn rate of 21.2%, compared with 19.7% among long-term customers. Credit-score segments showed comparatively smaller differences, indicating that credit score may be more useful as a supporting segmentation factor than as a primary indicator of churn risk in this analysis.

Based on these findings, the project recommends a segment-specific customer retention approach rather than a uniform retention strategy. Priority attention should be given to high-risk customer groups, particularly customers aged 46–60, customers in Germany, inactive customers, and high-balance customers. Recommended actions include targeted retention campaigns, proactive re-engagement of inactive customers, strengthened onboarding and early-stage engagement, personalized relationship management for high-value customers, and periodic monitoring of churn indicators through the Tableau dashboard.

Overall, the analysis demonstrates how customer segmentation and churn analytics can help banking institutions identify where customer attrition is concentrated and prioritize retention resources more effectively. The developed analytical framework and Tableau dashboard provide a practical foundation for data-driven customer management and can be extended in the future through richer behavioural data and predictive churn modelling.

ABSTRACT

Customer churn is a major challenge for banking institutions because the loss of existing customers can reduce customer lifetime value, increase customer acquisition costs, and affect long-term business stability. This project, titled “Customer Segmentation & Churn Pattern Analytics in European Banking,” focuses on identifying customer segments that demonstrate higher churn risk and understanding the key patterns associated with customer attrition.

The analysis uses customer-level banking data containing demographic, geographic, financial, product usage, tenure, and engagement-related variables. Exploratory Data Analysis (EDA) and customer segmentation techniques were used to examine churn patterns across geography, age groups, credit scores, tenure, account balance, and customer activity. Additional customer segments were created to support a more meaningful comparison of churn behaviour.

The findings highlight significant differences in churn across customer segments and indicate that certain demographic, geographic, financial, and engagement groups require greater retention attention. The analysis is presented through an interactive Tableau dashboard to support visual exploration and business decision-making.

Based on the identified patterns, targeted customer retention and re-engagement strategies are recommended for higher-risk segments. The overall objective of the project is to transform customer data into actionable business insights that can help banking institutions improve retention, protect customer lifetime value, and develop more targeted customer management strategies.

Keywords: Customer Churn, Customer Segmentation, Banking Analytics, Exploratory Data Analysis, Customer Retention, Business Analytics

1. INTRODUCTION

Customer retention is an important factor in the banking industry because maintaining existing customers is generally more sustainable than continuously replacing customers who leave. Customer churn refers to the situation in which a customer stops using or maintaining a relationship with a banking institution. A high level of customer churn can negatively affect revenue stability, customer lifetime value, and the overall efficiency of customer acquisition efforts.

The banking industry serves customers with different demographic, geographic, financial, and behavioural characteristics. As a result, customers do not demonstrate the same level of churn risk. Some customer groups may remain highly engaged with the bank, while others may show lower activity or stronger indicators of potential attrition. Identifying these differences is important for developing targeted customer retention strategies.

Customer segmentation provides a structured approach for understanding these differences. By dividing customers into meaningful groups based on characteristics such as geography, age, credit score, account balance, tenure, and activity status, organizations can identify segments with relatively higher churn rates. This allows decision-makers to move beyond overall churn measurement and focus on specific customer groups that may require intervention.

This project, “Customer Segmentation & Churn Pattern Analytics in European Banking,” applies exploratory data analysis and customer segmentation to identify patterns in customer churn. The analysis examines how churn varies across different customer characteristics and financial or engagement-related segments. The results are presented through an interactive Tableau dashboard to support clear visualization and interpretation of the findings.

The primary purpose of this analysis is not only to measure customer churn but also to identify where churn is concentrated and translate those findings into practical business recommendations. The insights generated through the analysis can help banking institutions develop more focused retention, re-engagement, and customer relationship strategies.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Customer churn represents a significant business challenge for banking institutions because the loss of existing customers can reduce customer lifetime value, increase the cost of acquiring replacement customers, and create instability in the customer base. However, an overall churn rate alone does not explain which types of customers are more likely to leave.

The key challenge addressed in this project is to identify customer segments with higher churn tendencies and understand the patterns associated with customer attrition. The analysis focuses on demographic, geographic, financial, tenure, and engagement-related characteristics to determine where churn is concentrated.

The project therefore aims to transform customer-level banking data into actionable insights that can support targeted customer retention strategies.

2.2 Objectives

The main objectives of the project are:

1. To measure the overall customer churn rate within the available banking dataset.
2. To identify differences in churn behaviour across geographical segments.
3. To analyse churn patterns across different age groups.
4. To examine the relationship between credit score and customer churn.
5. To analyse the impact of customer tenure on churn behaviour.
6. To identify churn patterns across different account balance groups.
7. To compare churn between active and inactive customers.
8. To identify high-risk and potentially high-value customer segments requiring greater retention attention.
9. To develop data-driven business recommendations for improving customer retention and reducing churn risk.

3. DATASET DESCRIPTION

3.1 Dataset Overview

The analysis is based on a customer-level banking dataset designed to study customer churn and segmentation patterns. The dataset contains demographic, geographic, financial, product-related, tenure, and customer engagement information.

The primary outcome variable used for the analysis is the customer churn indicator, represented by the variable **Exited**. This variable indicates whether a customer has left the banking institution or remained with the institution.

The dataset provides sufficient customer-level information to compare churn behaviour across different customer segments and identify groups that demonstrate relatively higher churn tendencies.

3.2 Variables Used in the Analysis

Variable	Description
Customer_Id	Unique identification number assigned to each customer
Credit Score	Credit score associated with the customer
Geography	Country/region of the customer
Gender	Gender of the customer
Age	Age of the customer
Tenure	Number of years the customer has been associated with the bank
Balance	Account balance maintained by the customer
NumOfProducts	Number of banking products held by the customer
Has Cr Card	Indicates whether the customer holds a credit card
Is Active Member	Indicates whether the customer is an active member
Estimated Salary	Estimated annual salary of the customer
Exited	Indicates whether the customer has churned

3.3 Target Variable

The variable **Exited** is treated as the primary churn indicator in this analysis.

- **Exited = 1:** Customer has churned.

- **Exited = 0:** Customer has remained with the bank.

The target variable is used throughout the analysis to calculate the overall churn rate and compare churn behaviour across different customer segments.

3.4 Analytical Segments

To make the analysis more meaningful from a business perspective, selected continuous variables were converted into customer segments. These include:

- Age Groups
- Credit Score Groups
- Tenure Groups
- Balance Groups
- Active and Inactive Customer Groups

These segments were subsequently used to compare churn rates and identify higher-risk customer categories.

4. RESEARCH METHODOLOGY

4.1 Analytical Approach

The project follows a structured Business Analytics approach to identify customer churn patterns and high-risk customer segments in the banking dataset. The methodology combines data understanding, data preparation, customer segmentation, exploratory data analysis, visualization, and business interpretation.

The overall analytical process followed in the project is:

Data Understanding → Data Validation & Cleaning → Data Preparation → Customer Segmentation → Exploratory Data Analysis → Churn Analysis → Visualization → Insight Generation → Business Recommendations

4.2 Data Understanding and Validation

The first stage involved understanding the structure and characteristics of the available customer dataset. The variables were reviewed based on their data types, business meaning, and relevance to customer churn.

The churn indicator, **Exited**, was examined to ensure that the target variable was appropriately represented as a binary variable. Relevant demographic, financial, tenure, product usage, and customer activity variables were also reviewed before conducting the analysis.

4.3 Data Cleaning and Preparation

The dataset was prepared before performing the analytical calculations. Unnecessary fields that did not contribute to the churn analysis were excluded from the analytical dataset.

The relevant variables were checked for consistency and suitability for analysis. Categorical and binary variables were prepared for comparison, while numerical variables were examined to support meaningful segmentation and visualization.

Derived analytical groups were also created for variables such as age, credit score, tenure, and account balance. These groups enabled the comparison of churn behaviour across clearly defined customer segments.

4.4 Customer Segmentation

Customer segmentation was performed to identify differences in churn behaviour among groups with similar characteristics.

The major segmentation dimensions used in the analysis were:

- **Geography:** Customer country
- **Age Group:** Customers grouped into defined age ranges
- **Credit Score Group:** Customers classified according to credit score ranges
- **Tenure Group:** Customers grouped according to their relationship duration with the bank
- **Balance Group:** Customers classified according to their account balance
- **Activity Status:** Active and inactive customers

These segments were analysed using churn rates rather than only customer counts, allowing meaningful comparisons between groups of different sizes.

4.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to identify patterns, differences, and potential risk segments within the customer base.

The analysis examined:

1. Overall customer churn rate
2. Churn rate by geography
3. Churn rate by age group
4. Churn rate by credit score group
5. Churn rate by tenure
6. Churn rate by balance group
7. Churn rate by customer activity status

The results were visualized using Tableau to make the patterns easier to interpret and compare.

4.6 Churn Rate Analysis

For segment-level comparison, churn rate was calculated as the proportion of customers who exited within a particular segment.

The general calculation used was:

$$\text{Churn Rate} = (\text{Number of Churned Customers} / \text{Total Customers in the Segment}) \times 100$$

This approach allows the analysis to identify segments with relatively higher churn risk instead of simply identifying segments with the largest number of churned customers.

4.7 Visualization and Insight Generation

The analytical results were presented through an interactive Tableau dashboard. Filters and visualizations were used to examine churn patterns across different customer characteristics.

The final stage involved interpreting the analytical results from a business perspective. The identified patterns were converted into actionable recommendations focused on customer retention, re-engagement, and targeted relationship management.

4.8 Business Interpretation

The methodology does not attempt to establish direct causal relationships between individual customer characteristics and churn. Instead, it focuses on identifying observable patterns and segments associated with higher or lower churn rates.

These findings are then used to support data-driven business recommendations for improving customer retention and prioritizing higher-risk customer segments

5. DATA PREPARATION AND CUSTOMER SEGMENTATION

5.1 Data Preparation

Before performing the churn analysis, the customer dataset was prepared to ensure that the available variables could be used effectively for segmentation and comparison.

The dataset was reviewed for variable consistency, appropriate data types, and the relevance of individual fields to the analytical objectives. Variables that were not required for the churn analysis were excluded from the analytical workflow.

The **Exited** variable was retained as the primary churn indicator, while demographic, geographic, financial, tenure, product, and engagement-related variables were used to understand differences in customer behaviour.

The prepared dataset was then used for exploratory analysis and customer segmentation.

5.2 Customer Segmentation Approach

Customer segmentation was performed to divide the customer base into meaningful groups and compare churn behaviour across those groups.

The segmentation dimensions used in the project are described below.

5.2.1 Age Group Segmentation

Customers were divided into defined age groups to identify whether churn behaviour differs across different stages of the customer life cycle.

Age-based segmentation helps identify age groups that may require different customer engagement and retention strategies.

5.2.2 Credit Score Segmentation

Customers were classified into credit score groups to examine whether customers with different credit profiles demonstrate different levels of churn.

This segmentation provides an additional financial-risk perspective to the churn analysis.

5.2.3 Tenure Segmentation

Customers were grouped according to the number of years they had been associated with the bank.

Tenure-based analysis helps determine whether churn is more concentrated among newer customers, established customers, or long-term customers.

5.2.4 Balance Segmentation

Customers were classified into balance groups based on their account balance.

This segmentation was used to identify whether customers with different financial value levels demonstrate different churn patterns and to highlight potentially high-value customer segments.

5.2.5 Activity Segmentation

Customers were divided into **Active** and **Inactive** groups using the customer activity indicator.

This segmentation was particularly important for identifying whether lower customer engagement is associated with higher churn.

5.2.6 Geography Segmentation

Customers were analysed according to their country of residence.

Geographical segmentation enables comparison of churn behaviour across different banking markets and helps identify countries where customer retention may require greater attention.

5.3 Purpose of Segmentation

The purpose of segmentation was to move beyond the overall churn rate and identify **where churn is concentrated within the customer base**.

By comparing churn rates across different customer segments, the analysis provides a clearer basis for identifying high-risk groups and developing targeted customer retention strategies.

6. EXPLORATORY DATA ANALYSIS AND RESULTS

The exploratory data analysis was conducted to examine customer churn patterns across key demographic, geographic, financial, tenure, and engagement-related segments. The analysis focuses on comparing segment-level churn rates to identify customer groups demonstrating relatively higher or lower churn tendencies. The findings are presented through statistical comparisons and Tableau visualizations to support clear interpretation and business decision-making.

6.1 Overall Customer Churn Analysis

The overall customer churn rate was analysed to establish a baseline for understanding customer attrition within the banking dataset. The analysis shows that **20.4% of the customers have churned**, while the remaining customers have continued their relationship with the bank.

An overall churn rate of 20.4% indicates that approximately one out of every five customers in the analysed customer base has exited the bank. This represents a meaningful level of customer attrition and highlights the importance of identifying the specific customer segments in which churn is more concentrated.

The overall churn rate serves as the baseline for the subsequent segmentation analysis. Churn rates across geography, age group, credit score, tenure, balance, and customer activity are compared against this overall level to identify relatively higher-risk customer segments.

Key Observation: The overall customer churn rate is **20.4%**.

Business Interpretation: Since approximately one-fifth of the analysed customers have churned, a broad retention strategy alone may not be sufficient. Segment-level analysis is required to identify the customer groups contributing disproportionately to churn and to support more targeted retention initiatives.

6.2 Geography-wise Churn Analysis

Customer churn was analysed across the three geographical segments in the dataset: France, Germany, and Spain. The analysis shows considerable variation in churn rates across countries.

Germany recorded the highest churn rate at **32.44%**, followed by Spain at **16.67%** and France at **16.15%**. Germany therefore shows a substantially higher level of customer attrition compared with both France and Spain.

Compared with the overall churn rate of **20.4%**, Germany's churn rate is significantly higher, indicating that the German customer segment represents a comparatively higher-risk market. In contrast, the churn rates observed in France and Spain are below the overall churn level.

Key Observation

Geography	Germany	Spain	France	Overall
Churn Rate	32.44%	16.67%	16.15%	20.40%

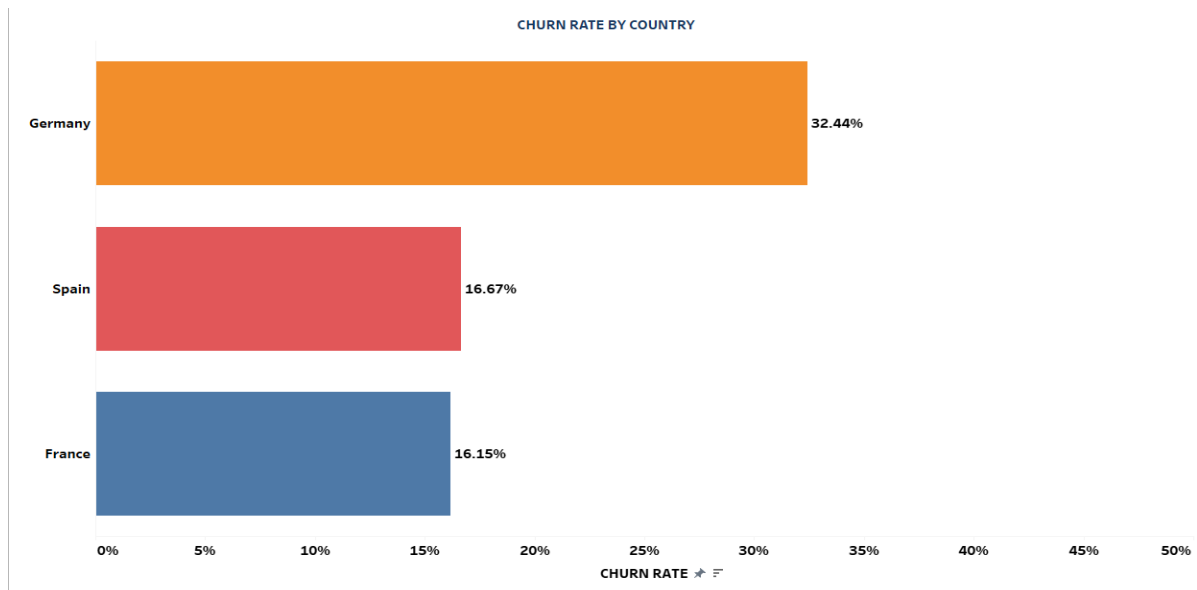


Figure 1: Geography-wise Customer Churn Rate

Interpretation

The significant difference in churn across countries indicates that customer attrition is not evenly distributed geographically. Germany requires greater attention from a customer-retention perspective because its churn rate is substantially above the overall level.

Business Implication

A country-specific retention strategy may be more effective than applying a uniform retention approach across all markets. The German customer segment should therefore be examined further using other characteristics such as age, balance, tenure, credit score, and activity status to identify the specific customer profiles contributing to the higher churn rate.

6.3 Age Group-wise Churn Analysis

Customer churn was analysed across different age groups to identify whether churn behaviour varies according to customer age. The analysis shows a substantial difference in churn rates across the defined age segments.

The churn rate was 7.5% among customers aged 18–30, 15.7% among customers aged 31–45, 51.1% among customers aged 46–60, and 24.8% among customers aged 61 and above.

The 46–60 age group recorded the highest churn rate at 51.1%, which is considerably higher than the overall churn rate of 20.4%. In comparison, customers aged 18–30 recorded the lowest churn rate at 7.5%.

Key Observation

Age Group	18–30	31–45	46–60	61+
Churn Rate	7.5%	15.7%	51.1%	24.8%

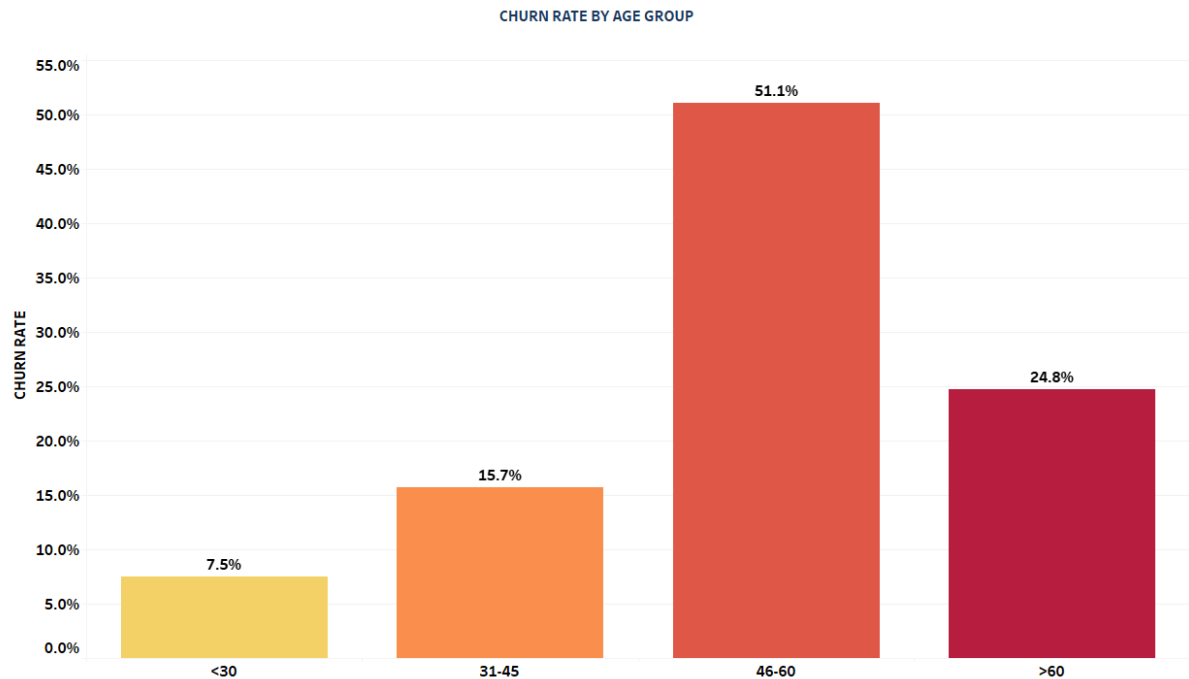


Figure 2: Age Group-wise Customer Churn Rate

Interpretation

The results indicate that churn is strongly concentrated in the 46–60 age segment. The substantially higher churn rate in this group suggests that age-related customer needs, expectations, financial preferences, or relationship factors may require further investigation.

Business Implication

The 46–60 customer segment should be treated as a high-priority retention segment. Banks can consider personalized relationship management, targeted retention offers, and proactive engagement strategies for customers within this age group.

6.4 Credit Score-wise Churn Analysis

Customer churn was analysed across credit score groups to determine whether churn behaviour differs according to customers' credit profiles.

The churn rate was **21.1%** for the Low credit score group, **18.8%** for the Medium credit score group, and **20.1%** for the High credit score group.

The Low credit score group recorded the highest churn rate at **21.1%**, while the Medium credit score group recorded the lowest churn rate at **18.8%**. The differences between the three groups are relatively small compared with the larger variations observed across other segmentation dimensions.

Key Observation

Credit Score Group	Low	Medium	High
Churn Rate	21.1%	18.8%	20.1%

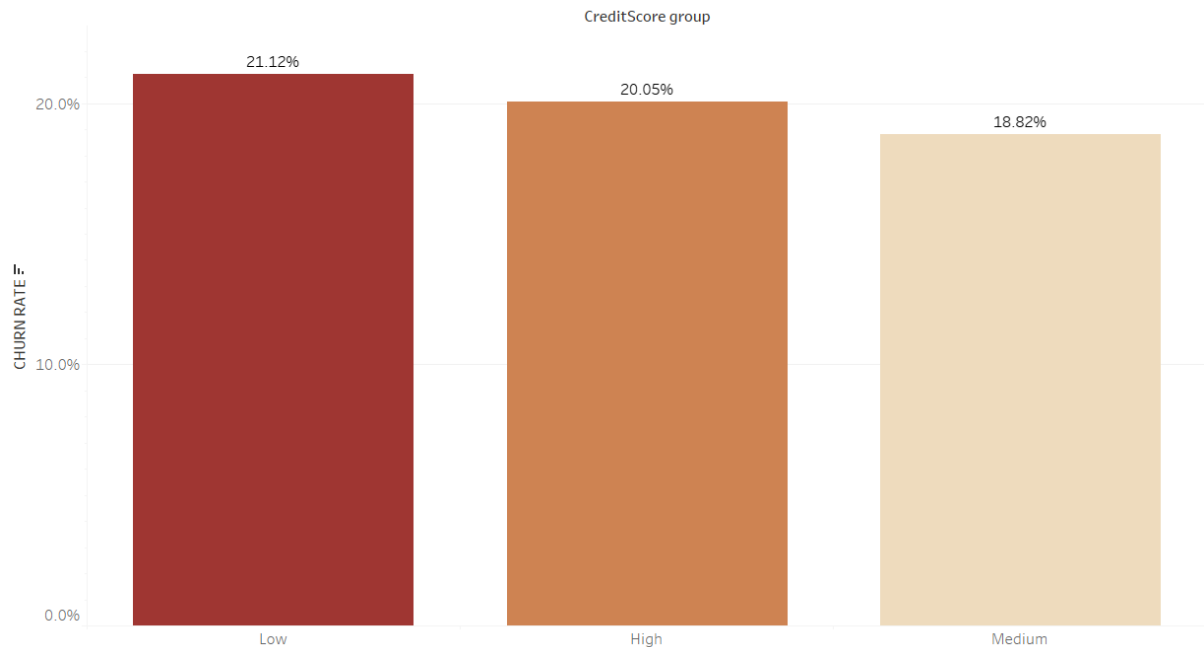


Figure 3: Credit Score Group-wise Customer Churn Rate

Interpretation

The results suggest that credit score does not show a very strong variation in churn compared with factors such as age or customer activity. However, the Low credit score segment has a slightly higher churn rate than the overall churn rate of 20.4%.

Business Implication

Credit score can be considered as a supporting factor in customer segmentation rather than the primary indicator of churn risk. Retention strategies should therefore consider credit score together with demographic, financial, tenure, and engagement characteristics.

6.5 Tenure-wise Churn Analysis

Customer tenure was analysed to understand whether the length of the customer's relationship with the bank is associated with differences in churn behaviour.

The churn rate was 21.2% among new customers, 20.8% among mid-term customers, and 19.7% among long-term customers.

New customers recorded the highest churn rate at 21.2%, while long-term customers recorded the lowest churn rate at 19.7%. Although the differences are relatively small, the results indicate a gradual decline in churn as customer tenure increases.

Key Observation

Tenure Group	New Customer	Mid-Term	Long-Term
Churn Rate	21.2%	20.8%	19.7%

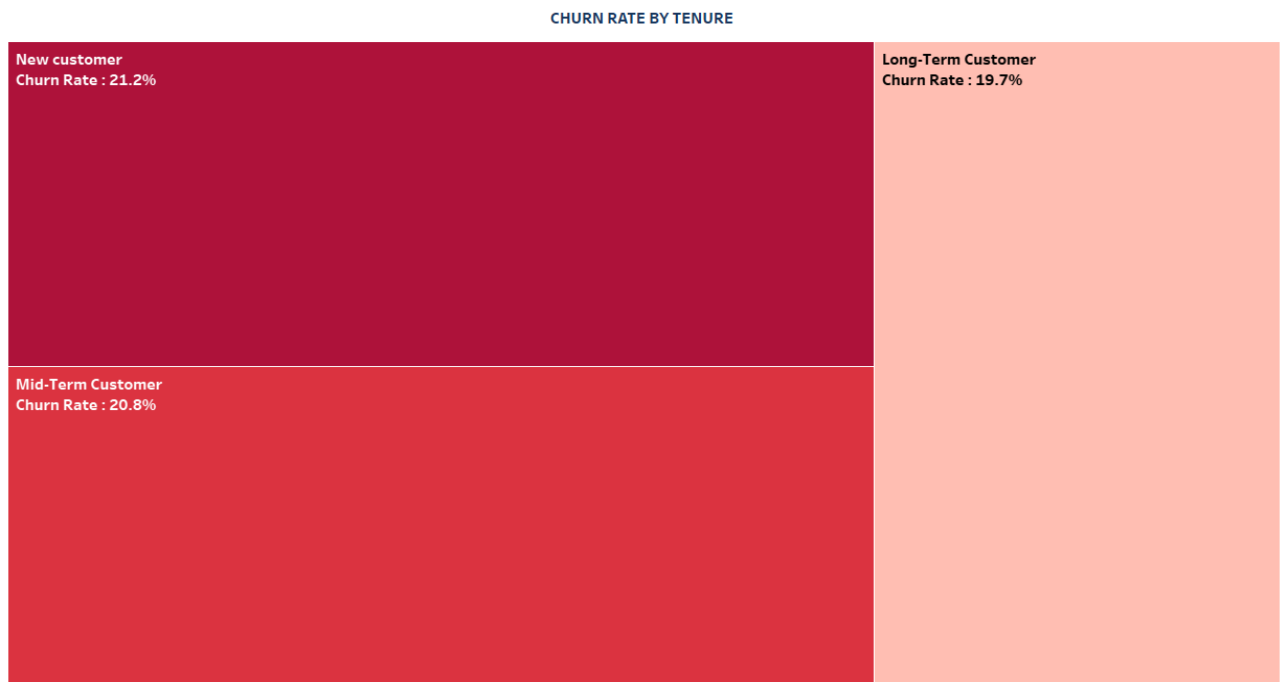


Figure 4: Tenure-wise Customer Churn Rate

Interpretation

The relatively higher churn among new customers may indicate that the early stage of the customer relationship is important for establishing engagement and satisfaction. Long-term customers show comparatively lower churn, suggesting that established relationships may provide greater customer stability.

Business Implication

Banks should focus on strengthening customer relationships during the early stages of the customer lifecycle. Effective onboarding, regular engagement, and early identification of disengaged customers may help reduce early-stage churn.

6.6 Balance Group-wise Churn Analysis

Customer churn was analysed across account balance groups to understand whether churn varies according to customers' financial value.

The churn rate was 14.25% for customers with zero balance, 24.11% for customers in the minimum/low balance group, and 23.12% for customers in the high balance group.

The Minimum/Low Balance group recorded the highest churn rate at 24.11%, followed by the High Balance group at 23.12%. Customers with zero balance recorded the lowest churn rate at 14.25%.

Key Observation

Balance Group	Zero Balance	Minimum/Low Balance	High Balance
Churn Rate	14.25%	24.11%	23.12%

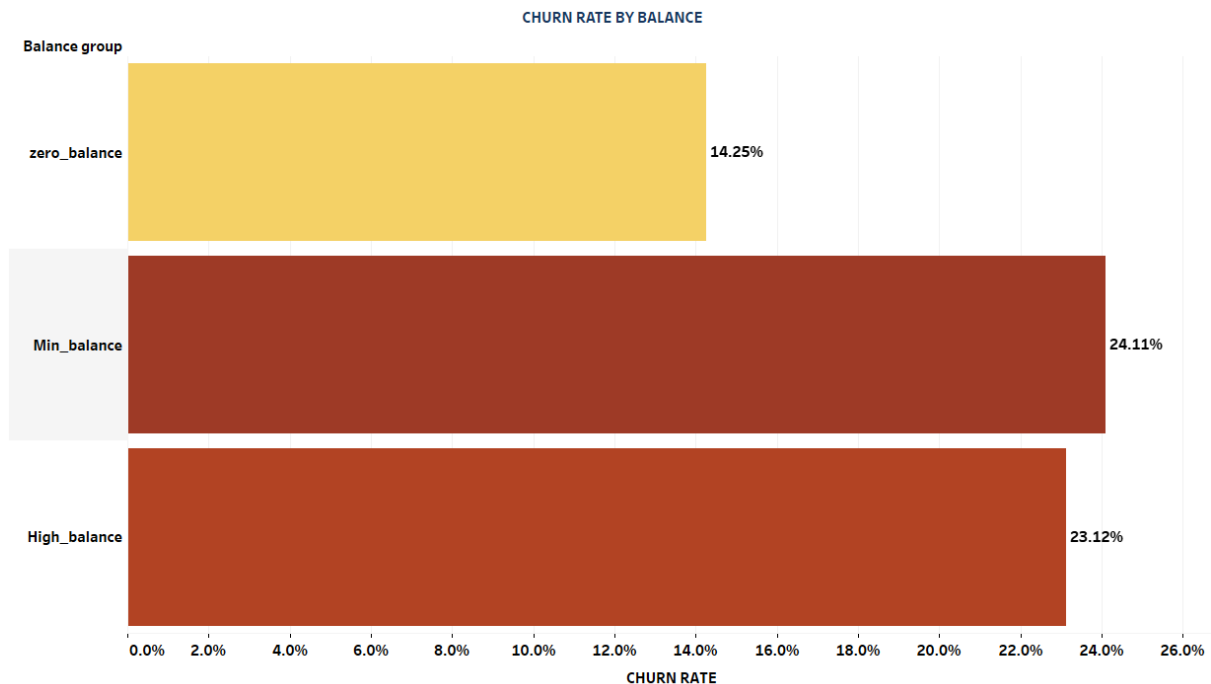


Figure 5: Balance Group-wise Customer Churn Rate

Interpretation

Customers in the Minimum/Low Balance and High Balance groups both show churn rates above the overall churn rate of 20.4%. This indicates that financial value alone does not provide a straightforward explanation of churn behaviour.

The relatively high churn rate among high-balance customers is particularly important from a business perspective because losing such customers may have a greater potential impact on the bank's customer value.

Business Implication

High-balance customers should receive proactive retention attention because their departure may represent a potentially significant loss of customer value. At the same time, the higher churn among low-balance customers suggests the need for differentiated engagement strategies across financial segments.

6.7 Active versus Inactive Customer Churn Analysis

Customer activity status was analysed to determine whether customer engagement is associated with differences in churn behaviour.

The analysis shows a churn rate of **14.27%** among Active customers compared with **26.85%** among Inactive customers.

Inactive customers therefore recorded a substantially higher churn rate than Active customers. The churn rate for Inactive customers is also considerably above the overall churn rate of 20.4%.

Key Observation

Activity Status	Churn Rate
Active	14.27%
Inactive	26.85%



Figure 6: Active vs Inactive Customer Churn Rate

Interpretation

The results indicate a clear difference in churn behaviour based on customer activity. Inactive customers demonstrate a substantially higher tendency to churn compared with active customers.

This makes customer activity one of the more prominent indicators of churn risk identified in the analysis. Lower engagement may therefore serve as an important signal for identifying customers who require proactive attention.

Business Implication

Banks should prioritize re-engagement strategies for inactive customers. Early identification of declining customer activity, followed by personalized communication, relevant product offers, and relationship-management initiatives, may help reduce the likelihood of customer churn.

7. KEY FINDINGS AND INSIGHTS

The exploratory analysis identified several important patterns in customer churn across geographical, demographic, financial, tenure, and engagement-related segments. The major findings are summarized below.

7.1 Overall Churn

The overall customer churn rate in the analysed dataset is **20.4%**. This indicates that approximately one in five customers has exited the bank, highlighting the importance of effective customer retention strategies.

7.2 Geographic Churn Concentration

Germany recorded the highest churn rate at **32.44%**, substantially above the overall churn rate of 20.4%. Spain and France recorded lower churn rates of **16.67%** and **16.15%**, respectively.

This indicates that customer churn is not evenly distributed across geographical markets and that Germany represents a comparatively higher-risk market.

7.3 Age-related Churn Risk

Age shows one of the strongest variations in churn behaviour. Customers aged **46–60 recorded the highest churn rate at 51.1%**, while customers aged 18–30 recorded the lowest rate at 7.5%.

The substantial difference suggests that age-based customer needs and engagement patterns should be considered when designing retention strategies.

7.4 Credit Score Shows Limited Variation

Churn rates across credit score groups were relatively close to each other. The Low credit score group recorded **21.1%**, High credit score customers recorded **20.1%**, and Medium credit score customers recorded **18.8%**.

Compared with age and activity status, credit score therefore shows relatively limited variation in churn within this dataset.

7.5 New Customers Show Slightly Higher Churn

New customers recorded a churn rate of **21.2%**, compared with **20.8%** for mid-term customers and **19.7%** for long-term customers.

The gradual reduction in churn across increasing tenure suggests that the early customer relationship stage may require greater attention to onboarding, engagement, and relationship development.

7.6 Balance-related Churn

The Minimum/Low Balance group recorded the highest churn rate at **24.11%**, followed by the High Balance group at **23.12%**. The Zero Balance group recorded a lower churn rate of **14.25%**.

The relatively high churn among High Balance customers is particularly important because these customers may represent greater potential customer value for the bank.

7.7 Customer Activity is a Strong Churn Indicator

A significant difference was observed between Active and Inactive customers. Active customers recorded a churn rate of **14.27%**, whereas Inactive customers recorded **26.85%**.

The substantially higher churn among inactive customers indicates that customer engagement is an important factor associated with churn risk and should receive particular attention in retention planning.

7.8 Overall Business Insight

The analysis indicates that churn risk is not driven equally by all customer characteristics. **Age, geography, and customer activity show particularly notable differences in churn rates**, while credit score and tenure demonstrate relatively smaller variations.

The findings support a segmented approach to customer retention, where higher-risk customer groups are identified and targeted with appropriate engagement and retention strategies rather than applying a uniform approach to the entire customer base.

8. BUSINESS RECOMMENDATIONS

The findings from the customer churn analysis indicate that a uniform retention strategy may not be equally effective across all customer segments. The bank can use the identified high-risk segments to develop targeted and proactive retention initiatives.

8.1 Targeted Retention Strategy for Germany

Germany recorded the highest churn rate at **32.44%**, significantly above the overall churn rate of 20.4%.

The bank should conduct a deeper segment-level analysis of German customers and identify the demographic, financial, tenure, and engagement characteristics associated with their higher churn. Country-specific retention campaigns, personalized communication, and proactive relationship management can then be developed for high-risk German customers.

8.2 Focus on Customers Aged 46–60

The 46–60 age group recorded the highest churn rate at **51.1%**.

This segment should receive priority retention attention. The bank can develop personalized financial solutions, relationship-management programs, and targeted communication based on the needs of this customer group.

8.3 Re-engagement of Inactive Customers

Inactive customers recorded a churn rate of **26.85%**, compared with 14.27% among active customers.

Customer inactivity can therefore be used as an early warning signal for potential churn. The bank should implement proactive re-engagement campaigns, including personalized communication, relevant product recommendations, digital engagement initiatives, and relationship-manager follow-ups.

8.4 Retention of High-Balance Customers

The High Balance segment recorded a churn rate of **23.12%**, which is above the overall churn rate of 20.4%.

Because high-balance customers may represent greater financial value to the bank, they should be monitored through a priority retention framework. Proactive relationship management and personalized service offerings can be used to strengthen customer relationships and reduce potential value loss.

8.5 Improve Early Customer Engagement

New customers recorded a churn rate of **21.2%**, slightly higher than the overall churn rate.

The bank should strengthen the early stages of the customer lifecycle through effective onboarding, regular communication, digital engagement, and early identification of disengaged customers. Building a strong relationship during the initial period may help improve long-term customer retention.

8.6 Differentiate Retention Strategies by Customer Segment

The analysis demonstrates that churn varies across geography, age, balance, and activity status. Therefore, the bank should avoid relying solely on a single retention campaign for the entire customer base.

Instead, customers can be prioritized according to their observed churn risk and customer value. This can support more efficient allocation of retention resources and more relevant customer communication.

8.7 Develop a Proactive Churn Monitoring Framework

The Tableau dashboard developed as part of this project can be used as a monitoring tool for regularly reviewing churn patterns across customer segments.

A periodic review of geography, age, balance, tenure, credit score, and activity-related churn metrics can help management identify changes in high-risk segments and support timely intervention.

8.8 Overall Strategic Recommendation

The overall findings support a **data-driven, segment-specific retention strategy** rather than a one-size-fits-all approach. Priority should be given to customers and segments showing both higher churn risk and greater potential business value.

By combining customer segmentation, engagement monitoring, and targeted retention actions, the bank can improve its ability to identify at-risk customers and allocate retention efforts more effectively.

9. LIMITATIONS AND FUTURE SCOPE

9.1 Limitations

Although the analysis provides useful insights into customer churn patterns, several limitations should be considered when interpreting the findings.

9.1.1 Dataset Limitation

The analysis is based on the variables available in the selected customer dataset. Factors such as customer satisfaction, complaints, service quality, competitor activity, transaction behaviour, and specific reasons for leaving the bank are not directly available. Therefore, the analysis cannot capture every factor that may influence customer churn.

9.1.2 Observational Nature of the Analysis

The analysis identifies associations and patterns between customer characteristics and churn. It does not establish that a particular customer characteristic directly causes churn. Therefore, the findings should be interpreted as analytical patterns rather than direct causal relationships.

9.1.3 Limited Behavioural Information

The dataset contains selected indicators of customer engagement, such as activity status, but does not provide detailed behavioural information such as transaction frequency, digital banking usage, customer complaints, or changes in customer activity over time.

9.1.4 Static Analysis

The analysis is based on the available customer records and provides a snapshot of churn patterns. It does not implement a time-series or predictive churn model to estimate how customer churn risk may change over time.

9.1.5 Generalizability

The findings are specific to the analysed dataset and its customer characteristics. Therefore, the results should not automatically be generalized to every banking institution or market without further validation using additional datasets.

9.2 Future Scope

Future research can extend this analysis by incorporating richer customer-level behavioural and transactional information.

Predictive machine learning models could be developed to estimate the probability of individual customers churning. Additional variables such as transaction frequency, digital engagement, customer complaints, product usage, and changes in account activity could also improve the identification of at-risk customers.

Future studies could further validate the identified high-risk segments using data from multiple periods, institutions, or geographical markets. This would help determine whether the observed churn patterns remain consistent across different customer populations.

The Tableau dashboard developed in this project can also be extended into a more advanced monitoring system that regularly tracks customer churn indicators and supports proactive identification of emerging high-risk segments.

Overall, future research can build upon the current descriptive and segmentation-based analysis by combining richer behavioural data, predictive analytics, and continuous churn monitoring.

10. CONCLUSION

This project analysed customer churn patterns in a European banking customer dataset using exploratory data analysis and customer segmentation techniques. The analysis examined churn behaviour across geographical, demographic, financial, tenure, and customer engagement characteristics to identify segments with comparatively higher churn risk.

The overall churn rate in the analysed customer base was **20.4%**. The analysis revealed substantial differences across several customer segments. Germany recorded the highest geographical churn rate at **32.44%**, while customers aged 46–60 recorded the highest age-group churn rate at **51.1%**. Customer activity also demonstrated a notable difference, with inactive customers recording a churn rate of **26.85%**, compared with **14.27%** among active customers.

Financial segmentation also identified important patterns. The Minimum/Low Balance group recorded a churn rate of **24.11%**, while the High Balance group recorded **23.12%**. In terms of tenure, new customers recorded a slightly higher churn rate of **21.2%**, compared with **19.7%** among long-term customers. Credit score groups showed comparatively smaller differences in churn rates.

Overall, the findings demonstrate that customer churn is not evenly distributed across the customer base. Certain segments, particularly customers aged 46–60, customers in Germany, inactive customers, and selected balance groups, require greater attention from a customer-retention perspective.

The analysis therefore supports the use of a **segment-based retention strategy** rather than a uniform approach for all customers. The Tableau dashboard developed as part of the project provides an interactive way to monitor these churn patterns and supports data-driven business decision-making.

Future development of this analysis could incorporate additional behavioural and transactional variables and predictive modelling techniques to identify individual customers with a higher

probability of future churn. Such extensions could help banking institutions move from descriptive churn analysis toward proactive and predictive customer retention.